Plant Systematics and Evolution (2026) 312:57
https://doi.org/10.1007/s00606-026-02027-7

RESEARCH

Phylogenomics of the holoparasitic plant genus *Hydnora* (Hydnoraceae)

Sebastian A. Hatt[1,2] · Alice Merrall[1] · Jay F. Bolin[3] · Lytton J. Musselman[4] · Duncan D. Cameron[2] · Chris J. Thorogood[5,6] · Alexandre R. Zuntini[1] · Olwen M. Grace[1,7]

Received: 27 March 2026 / Accepted: 10 August 2026
© The Author(s) 2026

Abstract

Hydnora is a highly unusual genus of holoparasitic plants in the family Hydnoraceae comprising 10 species across Africa and the Middle East. Species concepts and relationships within the genus have historically been poorly defined, in part due to a paucity of morphological characters available for species delimitation. Here, we present the most comprehensively sampled nuclear phylogenomic study of *Hydnora* to date. We employ the Angiosperms353 target capture approach, using 34 nuclear loci to generate the species tree. Our results provide strong molecular support for all four recently proposed subgenera of *Hydnora*, demonstrating that even a limited set of morphological traits can be used to delimit lineages within the genus. We estimate ancestral states for these taxonomically informative characters within the Hydnoraceae and provide molecular evidence to clarify some of the long-standing taxonomic uncertainty within the widespread *Hydnora abyssinica* complex. We analyse gene tree discordance in the results and discuss our results in the context of these conflicting studies. The phylogenetic hypotheses presented here are the most widely sampled to date, and provide a framework for future ecological and evolutionary research into this enigmatic family of parasitic plants.

Keywords Angiosperms353 · Hydnora · Parasitism · Piperales · Phylogenetics · Taxonomy

Introduction

Hydnora Thunb. is a holoparasitic plant genus in the family Hydnoraceae with a highly specialised life history and morphology unusual amongst angiosperms (Hatt et al. 2023, 2024b) (Fig. 1). Lacking photosynthetic leaves or stems, *Hydnora* form branching, serpentiform rhizomes. Large fleshy flowers develop from meristematic warts on the rhizome and breach the soil surface during anthesis, attracting pollinators into the floral chamber through a combination of thermogenesis and emission of volatile organic compounds (Bolin et al. 2009a; Rocamundi et al. 2024). They are entirely dependent on a host plant for survival, deriving all nutrients from haustorial connections first established with the host root shortly after germination (Tennakoon et al. 2007; Bolin et al. 2009b). They are host specific, typically restricted to either a single host species, genus or family (Hatt et al. 2024a).

A recent taxonomic revision of *Hydnora* (Hatt et al. 2024b) recognised 10 species across Africa, Madagascar and southern Arabia (Table 1). *Hydnora* subgenus *Hydnora*, comprising four species in Southern Africa, parasitise succulent *Euphorbia* L. (Euphorbiaceae) hosts. The single East African species in subg. *Sineseta* S.Hatt parasitises *Commiphora* Jacq. (Burseraceae). Species in subg. *Dorhyna* Decne. (widespread) and subg. *Neohydnora* Harms (native to Madagascar) parasitise various tree species in Fabaceae.

Handling Editor: Karol Marhold.

✉ Sebastian A. Hatt
s.hatt@kew.org

1 Royal Botanic Gardens, Kew, London TW9 3AE, UK

2 Earth and Environmental Sciences, University of Manchester, Manchester, UK

3 Department of Biology, Catawba College, Salisbury, NC 28144, USA

4 Department of Biological Sciences, Old Dominion University, Norfolk, VA 23529, USA

5 Department of Biology, University of Oxford, Oxford OX1 3RB, UK

6 University of Oxford Botanic Garden and Arboretum, Oxford OX1 4AZ, UK

7 Royal Botanic Garden Edinburgh, Edinburgh EH3 5LR, UK

Published online: 09 October 2026

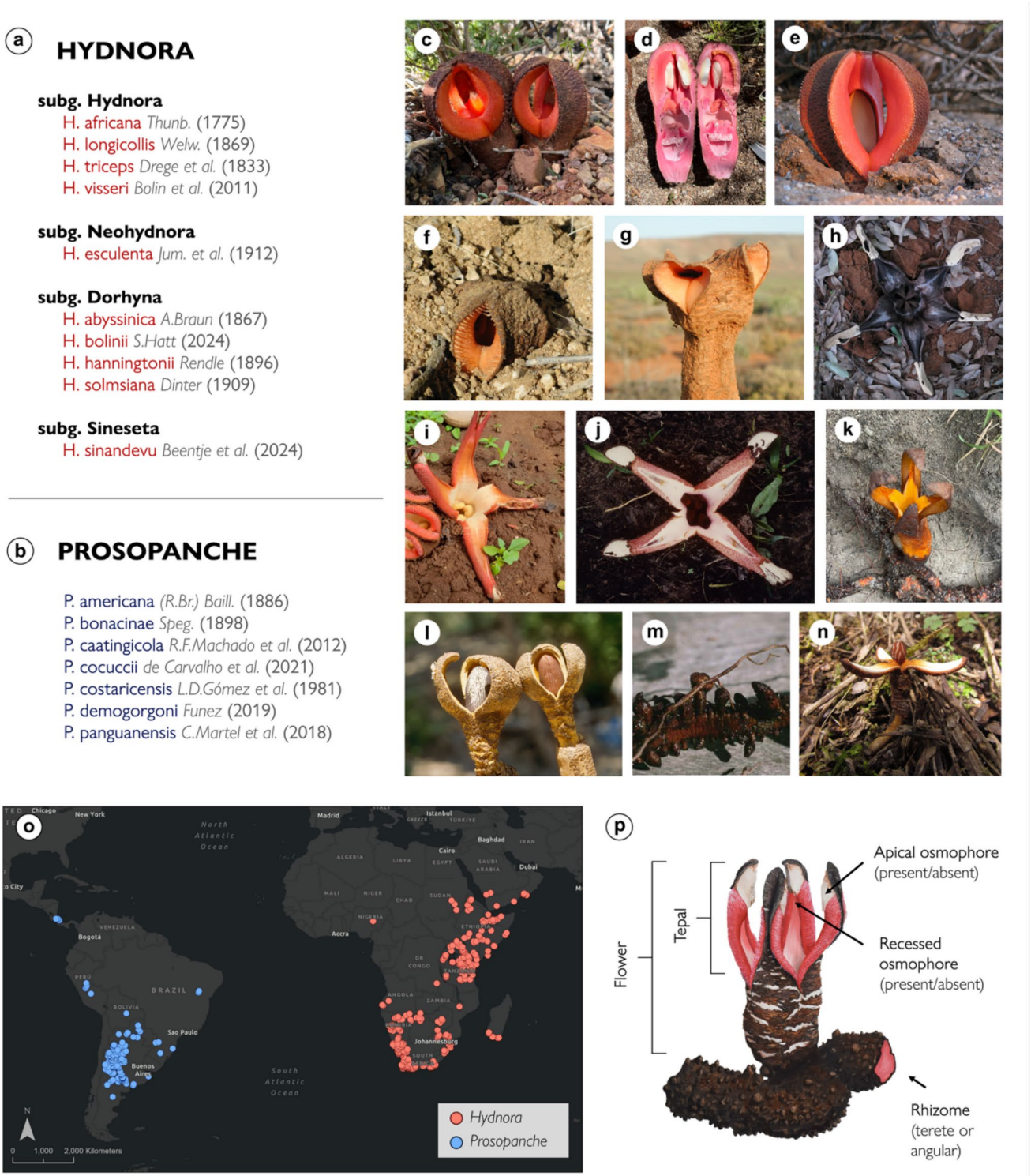


Fig. 1 Hydnoraceae. **a** Classification of *Hydnora*, following Hatt et al. (2024a, b). **b** Classification of *Prosopanche*, in alphabetical order. **c**–**g** *Hydnora* subg. *Hydnora*. **c** *Hydnora africana* in flower; **d** *H. africana* flower dissected; **e** *H. visseri* in flower; **f** *H. longicollis* in flower; **g** *H. triceps* in flower. **h** *Hydnora* subg. *Neohydnora*, *H. esculenta*. **i**–**j** *Hydnora* subg. *Dorhyna*. **i** *H. abyssinica* in flower; **j** *H. solmsiana* in flower. **k** *Hydnora* subg. *Sineseta*, *H. sinandevu*. **l**–**n** *Prosopanche*. **l** *Prosopanche americana* in flower; **m** *P. americana* rhizome and host root; **n** *P. demogorgoni*. **o** Distribution of Hydnoraceae. **p** Key taxonomic characters of a typical Hydnoraceae species, in this example *Hydnora abyssinica*. (Images used with permission from: **c**–**d**, Sebastian A. Hatt; **e**–**h**, Jay F. Bolin; **i** Matthew Rees; **j** from *Malaisse 11,290* [BR]; **k** Rob Barbour; **l**–**m** Lytton Musselman; **n** Luis Adriano Funez.). **p** illustrated by Sebastian A. Hatt

Table 1 A summary of characters all recognised 17 species of Hydnoraceae. Rhizome: *A* angular, *R* terete. Osmophore: *A* apical, *R* recessed. Informed by Hatt et al. (2024b)

Species	Clade	Rhizome	Osmophore	Host Specificity	Distribution
Hydnora africana	subg. Hydnora	A	R	Euphorbia (Euphorbiaceae)	Southern Africa
Hydnora visseri		A	R		
Hydnora longicollis		A	R		
Hydnora triceps		A	R		
Hydnora esculenta	subg. Neohydnora	A	A	Mimosoideae (Fabaceae)	Madagascar
Hydnora abyssinica	subg. Dorhyna	T	A	Mimosoideae (Fabaceae)	Southern to East Africa
Hydnora hanningtonii		T	A		East Africa to Oman
Hydnora solmsiana		T	A+R		Southern to East Africa
Hydnora bolinii		T	A+R		East Africa
Hydnora sinandevu	subg. Sineseta	T	N	Commiphora (Burseraceae)	East Africa to Oman
Prosopanche americana	n/a	A	A	Prosopis (Fabaceae)	Argentina
Prosopanche bonacinae	n/a	A	A	8 families (e.g. Asteraceae, Malvaceae, Solanaceae)	Argentina to Bolivia
Prosopanche caatingicola	n/a	A	A	Mimosoideae (Fabaceae)	Brazil (Caatinga)
Prosopanche cocuccii	n/a	A	A	Nicotiana (Solanaceae)	Brazil (South)
Prosopanche costaricensis	n/a	A	A	Mimosoideae (Fabaceae)	Costa Rica
Prosopanche demogorgonii	n/a	A	A	Baccharis (Asteraceae)	Brazil (South)
Prosopanche panguanensis	n/a	A	A	Mimosoideae (Fabaceae)	Peru

Several species are used in traditional medicine (Williams et al. 2011; Mkala et al. 2021), and their subterranean fruits are eaten by humans and animals (Hatt et al. 2024b). The only other genus in Hydnoraceae is *Prosopanche*, which comprises seven species (Hatt et al. 2023) distributed in South America and Costa Rica, three of which have been described in the last ten years (Martel et al. 2018; Funez et al. 2019; De Carvalho et al. 2021). Host preference is less well studied in *Prosopanche*, but preferred host families include Fabaceae, Asteraceae and Solanaceae (Cocucci 1965; Hatt et al. 2023, 2024a).

Taxonomic resolution of both genera has historically been poor, in part because of scarcity of material (Teixeira-Costa et al. 2023), their elusive flowering behaviour, recalcitrance to cultivation, and the paucity of morphological characters for species delimitation (Musselman and Visser 1989; Bolin et al. 2018; Hatt et al. 2024b). The use of molecular evidence to resolve species boundaries has also been challenging (Bolin 2009). Without the biochemical and physiological constraints imposed by photosynthesis, parasitic plant genomes are characteristically divergent from those of photosynthetic plants because of relaxed selection pressures, particularly in the plastome (dePamphilis and Palmer 1990; Gruzdev et al. 2019; Kim et al. 2026).

In his doctoral thesis, Bolin (2009) presented the first molecular phylogenetic tree of the genus, based on ITS and *rpoB* regions, with sequences derived predominantly from the southern African *Hydnora* subg. *Hydnora*, but also four other species from the genus and three from *Prosopanche*. Although clades within *Hydnora* generally lacked strong support, *H. esculenta* Jum. and H. Perrier was resolved as sister to all other *Hydnora*, and the two genera resolved as sister to each other. Jost et al. (2022) and Mkala et al. (2023) investigated plastome diversity using 83 plastid loci to reconstruct a phylogenetic tree. The latter study mainly sampled several very closely related individuals of *H. abyssinica* A.Br. from Kenya, but both studies included representatives from six other species. The topologies were largely congruent with that of Bolin (2009).

Most phylogenetic work involving *Hydnora* has focused on determining the placement of the family Hydnoraceae. Jost et al. (2021) used 83 plastid loci, 44 mitochondrial loci and 10 nuclear loci to investigate the relationship between Hydnoraceae and other Piperales. Similar to previous morphological observations (Meyer 1833) and phylogenetic studies (Nickrent et al. 2002; Naumann et al. 2013; Massoni et al. 2014), they found Hydnoraceae is likely to be nested within Piperales, sister to Aristolochiaceae sensu stricto (the circumscription that includes only *Aristolochia* and *Thottea*, excluding *Asarum*, *Lactoris*, and *Saruma*). More recently, a large-scale angiosperm-wide phylogenetic tree generated using 353 nuclear loci (Zuntini et al. 2024) positioned *Hydnora* as sister to all Piperales, a placement also recovered in a phylogenetic study of Magnoliids (Helmstetter et al. 2025) using the same 353 loci. However, in both studies, only a substantially reduced subset of the 353 target loci was successfully recovered from Hydnoraceae, with 28 and 84 genes retrieved from the two *Hydnora* samples, respectively.

Despite extensive alpha taxonomic work on the genus *Hydnora* conducted using morphological and ecological traits (Hatt et al. 2024b), molecular phylogenetic studies with broad taxon sampling have been lacking to test and inform the proposed evolutionary relationships within the genus. Here, we sample widely across *Hydnora* and apply target capture sequencing of up to 353 nuclear loci using the Angiosperms353 bait kit (Johnson et al. 2019). Given the difficulty of obtaining fresh material for sampling, herbarium specimens, many of which are over 40 years old, represented the only practical option for broad sequencing across the genus. The Angiosperms353 approach (Johnson et al. 2019) is widely regarded as one of the most effective methods for generating informative data from DNA extracted from herbarium material (Baker et al. 2021). As such, we use these data to test the species concepts and sectional classification of Hatt et al. (2024b), and infer ancestral character states for *Hydnora*.

Materials and methods

Distribution maps

To illustrate the known distribution of Hydnoraceae (Fig. 1), occurrence was compiled from herbarium specimens, literature and downloaded data points from GBIF (GBIF.org 2025), available here: https://doi.org/10.15468/dl.gp8k3r. The map was generated with these data using ArcGIS Pro 2.8 (ESRI 2026).

Taxon and tissue sampling

A total of 53 samples were obtained for phylogenomic analysis (Online Resource 1). Ten outgroup species were sampled, including nine *Prosopanche* samples and *Aristolochia littoralis* L. (Aristolochiaceae *s.l.*). The remaining 43 samples were of *Hydnora*. Taxa were selected to maximise taxonomic and geographic breadth. Given the uncertainty around species boundaries in *Hydnora* subg. *Dorhyna* (Hatt et al. 2024b), a particular effort was made to sample across the full extent of the geographic range of this subgenus. At least two samples were obtained for all ten species of *Hydnora* and four of the seven species of *Prosopanche*. Given the absence of leaf tissue, material for sequencing was sampled from the flower. This was favoured over rhizome tissue, which was typically heavily lignified and difficult to break down. Data for 47 samples were newly generated for this study, from material sourced from both herbarium specimens at K and MO (32 samples) and silica gel-dried field collections (15 samples). Material from these samples was used for DNA extraction, amplification and sequencing. The outgroup sample and five of the Hydnoraceae samples were retrieved in the form of raw reads downloaded from the European Nucleotide Archive (ENA, https://www.ebi.ac.uk/ena/browser/home). These reads were sequenced for the Plants and Fungal Trees of Life (PAFTOL) project (Baker et al. 2022) using the Angiosperms353 probe set (Johnson et al. 2019). A summary of all samples and their associated metadata is available in Online Resource 1.

DNA extraction, amplification and sequencing

DNA extraction following the CTAB protocol was described in Doyle and Doyle (1987). Quantus Fluorometer (Promega, Madison, WI, USA) measurements and gel electrophoresis using a 1% agarose gel were conducted to estimate extraction concentration and DNA fragment size. Library preparation, hybridisation with Angiosperms353 and sequencing were performed by Neogen (Lansing, MI, USA). DNA was fragmented enzymatically to a size of approximately 500 bp. Libraries were prepared with Kapa DNA Prep Kit (Illumina Inc., San Diego, CA, USA) according to the manufacturer's instructions. Neogen used a miniaturised Kapa PCR reaction (6 cycles). Pair-end libraries were analysed for size distribution using the TapeStation and quantified using Qubit. Libraries were pooled equimolar and assessed for quality with Fragment Analyzer (Agilent). Hybrid capture with Angiosperms353 probes (Johnson et al. 2019) was performed in pools of 24 samples for 16 h at 65 °C. Sequencing was performed at the NovaSeq 6000 platform at PE150 (2×150-bp paired-end reads). All newly generated raw reads are available at ENA (PRJEB to be added on publication).

Read assembly, cleaning and loci recovery

Read processing and loci assembly used CAPTUS 1.0.0 (Ortiz et al. 2023), a toolkit designed for the assembly of phylogenomic datasets from high-throughput sequencing. Reads were cleaned using *captus_assembly clean* to trim adaptor regions and assess read quality, using default parameters. The cleaned reads were then assembled de novo with MEGAHIT (Li et al. 2015) using *captus_assembly assemble*. Angiosperms353 target loci were extracted from the assembled reads with Scipio (Keller et al. 2008) using *captus_assembly extract*, with the Angiosperms353 target file (Johnson et al. 2019) as the reference set. This command generates a summary spreadsheet (Online Resource 2) that reports recovery statistics for all target loci for all reads. Recovery per locus per sample is summarised as a heatmap in Online Resource 3.

The dataset was filtered to include only Hydnoraceae samples with >25 recovered genes (Online Resource 1, 2). This threshold was selected after testing various cut-offs, representing a compromise between maximising sample

inclusion and maintaining data quality for phylogenetic inference.

Gene and species tree estimation

Extracted sequences of exons were aligned separately using *captus_assembly align* (Ortiz et al. 2023), which uses MAFFT v.7.526 (Katoh and Standley 2013). Paralogues were removed at this stage, as advised by the developers of CAPTUS, using the 'informed' option for *filter_method*. This meant that for each sample, the copy most similar to the reference sequence that was most frequently selected across all other samples in the alignment was retained, irrespective of its hit ranking. Paralogues were detected in 86 of the recovered loci, affecting 36 samples. The number of detected paralogues per locus ranged from 2 to 84 (mean = 6), with most loci containing a single copy across all samples (Online Resource 2). Alignments were scanned for site variability (Online Resource 4, 5) using the *summary* function from AMAS (Borowiec 2016).

Maximum likelihood gene trees were generated (Online Resource 6) using RAxML-NG v.1.2.2 (Kozlov et al. 2019) with default parameters recommended by the developers, including the nucleotide substitution model: LG+G8+F (recommended for its ability to handle fast and uneven rates of evolution expected in parasitic plants), and bootstrap replicates set to 500 for efficiency. Nodes were collapsed that had ≤ 15% bootstrap support (BS). This was performed using *nw_ed* from the Newick Utilities v.1.6 (Junier and Zdobnov 2010). Excessively long branches, often indicative of erroneous sequences, were detected and removed from each gene tree using TreeShrink v.1.3.9 (Mai and Mirarab 2018) with default parameters. This step was also included because of the well-documented occurrence of long-branch attraction in parasitic plants (Nickrent et al. 2004). Taxa flagged as outliers by TreeShrink were excluded from the initial gene alignments. Gene trees were re-estimated from the revised alignments using the same parameters as above. Species trees were then inferred from the unrooted gene trees using ASTRAL-III v.5.7.8 (Zhang et al. 2018). Two iterations were run, first with the *-t2* flag to assess branch support using local posterior probabilities (PP) and quartet scores, second with the *-t10* flag for use in DiscoVista analysis.

The maximum likelihood (ML) tree (Online Resource 7) was generated separately using the alignments and gene trees. Alignments, together with the corresponding gene trees, were trimmed using TreeShrink v.1.3.9 (Mai and Mirarab 2018). Trimmed gene alignments were concatenated into Phylip output format for RAxML using AMAS (Borowiec 2016). The ML tree was then generated from the concatenated alignments using RAxML-NG v.1.2.2 (Kozlov et al. 2019) with 100 bootstrap replicates. All trees were plotted in R with ape 5.0 (Paradis and Schliep 2019) and ggtree 3.0.2 (Yu et al. 2017).

To assess how phylogenomic inference was affected by using a subset of genes present in the highest number of Hydnoraceae samples, the analyses were re-run with different presence-absence thresholds visualised in Online Resource 8. The 40% threshold, resulting in 34 genes, was ultimately selected as it was found to yield the highest support values. This threshold was determined as a compromise between maximising the total number of genes analysed and ensuring sufficient overlap of informative genes across samples.

Computationally intensive bioinformatic analyses were conducted in the UKCropDiversity-HPC (Percival-Alwyn et al. 2024).

Assessing gene tree discordance

DiscoVista (Sayyari et al. 2018) was used to visualise and analyse discordance in gene trees at particular nodes of interest (Fig. 3). Annotation and clade definition files were generated that outlined clades of interests. These files were inputted first with mode '1' to yield the number and proportion of gene trees that supported the monophyly of a given clade and second with mode '5' to yield the relative frequency of all three alternative topologies at focal nodes.

To analyse gene tree conflict more closely, a phylogenetic 'Super Network' was generated (Online Resource 9) the combined gene trees using the SplitsTree App (Huson and Bryant 2024).

Ancestral state estimation

Hydnoraceae trait data were gathered from the most recent taxonomic monographs of *Hydnora* (Hatt et al. 2024b) and *Prosopanche* (Cocucci 1965). Four taxonomically informative characters were selected: rhizome cross section, apical and recessed osmophore presence, and host preference. These characters were scored in a matrix and incorporated into a simplified version tree that contained only one tip for each species (Fig. 4). Ancestral state estimation analysis on these characters was performed using the R package corHMM (Beaulieu et al. 2013) on a simplified version of the ASTRAL species tree that included only one representative sample per taxon. Nodes with bootstrap support values below 75% were collapsed into polytomies post-analysis. *Hydnora abyssinica* and *H. hanningtonii* were retained as separate tips despite their poor resolution in the species tree, as this better reflects the morphological evidence supporting their delimitation. The observed lack of resolution may be attributable to recent diversification not fully captured by the loci sampled in this study (see Discussion). Equal rates (ER), symmetrical (SYM), and all rates different (ARD) models

were compared using corrected Akaike information criterion (AICc) scores. For rhizome cross section, apical osmophore, and recessed osmophore traits, ER and SYM models showed equivalent fit (ΔAICc < 2), whereas the ARD model was consistently unsupported. For host preference, the ER model was clearly favoured (ΔAICc > 6 relative to SYM). Given the comparable performance of ER and SYM across traits and the increased parameter complexity of SYM, the ER model was selected for all ancestral state reconstructions. Posterior probabilities for ancestral states at each node were plotted as pie charts using ggtree 3.0.2. (Yu et al. 2017).

Results

Sequence capture recovery

DNA extracts (Online Resource 1) varied considerably in concentration. Herbarium specimen-derived samples typically performed lower than those from silica-dried material (Online Resource 10). Angiosperms353 gene recovery was relatively low for all samples of Hydnoraceae, ranging from 1 to 130 genes (mean = 40.5 genes). The sum of coding sequences per Hydnoraceae sample ranged from 189 to 46,599 base pairs (mean = 13,112 bp). In contrast, gene recovery for the *Aristolochia littoralis* sample was considerably higher (344 genes), with a higher total coding sequence length (429,319 bp). Recovery rates are reported in Online Resource 2 and Online Resource 10 and are summarised in a heat map (Online Resource 3). From the original 52 Hydnoraceae samples, 23 exceeded the established recovery threshold of >25 genes and were retained in the final species-level trees. These 23 samples comprised at least one representative from all 10 species of *Hydnora*. However, unfortunately *Prosopanche* is represented by only a single sample of *P. americana* (R.Br.) Baill. The mean proportion of variable sites per alignment for the 34 genes was 0.371 and ranged from 0.241 to 0.494 (Online Resource 4, 5). Due to weak support within *Hydnora* subg. *Hydnora*, we examined the proportion of variable sites in alignments restricted to sequences from this subgenus, which was found to be only 0.0565.

Relationships within Hydnoraceae

Hydnoraceae is resolved as monophyletic with maximum support, with *Prosopanche* consistently sister to *Hydnora* (Figs. 2, 3). The sole species of *Hydnora* subg. *Neohydnora*, *Hydnora esculenta*, is resolved as sister to all other *Hydnora* species. This position is fully supported in the ML tree (Online Resource 7) but receives lower support (78 PP) in the ASTRAL tree. Despite this relatively low support, the monophyly of that clade is strongly supported by 20 genes, while only strongly rejected by one, and alternative topologies have very low support (Fig. 3; Online Resource 11).

Hydnora subgenus *Hydnora* is consistently recovered as sister to a clade comprising *H.* subg. *Dorhyna* and *H.* subg. *Sineseta*, with maximum support in the ML tree (Online Resource 7) and strong support (> 93 BS) in ASTRAL tree (Figs. 2, 3). The monophyly of this clade in the ASTRAL tree is strongly supported by three gene trees but strongly rejected by four (Fig. 3; Online Resource 11). However, the primary topology received 48% support while the second and third alternative topologies received 32% and 20%, respectively.

Within *Hydnora* subg. *Hydnora*, *H. triceps* Drège and E.Mey is very poorly supported as sister to the *H. africana* complex (*H. africana*, *H. longicollis* Welw. and *H. visseri* Bolin, E.Maass and Musselman) in the ASTRAL tree (26 PP) (Fig. 2). This clade is strongly supported by 15 gene trees and only strongly rejected by three (Fig. 3; Online Resource 11). However, *H. triceps* does not resolve as sister in the ML tree (Online Resource 7) and there is no support for these nodes. Relationships within the *H. africana* complex are poorly supported in both topologies, essentially forming a polytomy. In particular, the monophyly of the *H. africana* complex is very poorly supported: only two genes trees strongly support it while seven strongly reject it (Fig. 3; Online Resource 11).

Hydnora sinandevu Beentje and Q.Luke, the sole species in subg. *Sineseta*, is resolved as sister to a monophyletic subg. *Dorhyna* with strong support in both ASTRAL (> 98 PP) and ML (100 BS) topologies. In the ASTRAL tree (Fig. 2), the topology at this node is strongly supported over alternative topologies (Fig. 3).

The ASTRAL tree strongly supports (100 PP) that subg. *Dorhyna* comprises two clades: Clade I (*H. bolinii* S.Hatt + *H. solmsiana*) and Clade II (*H. abyssinica* + *H. hanningtonii* Rendle). In the ASTRAL tree (Fig. 2), the topology at this node is strongly supported over alternative topologies (Fig. 3). However, while the ML tree (100 BS) (Online Resource 7) maintains the monophyly of Clade I with strong support, relationships between *H. abyssinica* and *H. hanningtonii* in Clade II are poorly resolved resulting in these two taxa being paraphyletic. Relationships between these two species are poorly resolved in the ASTRAL tree (Fig. 2), essentially forming a polytomy. The monophyly of Clade II is strongly supported by three genes, but also strongly rejected by three genes (Fig. 3; Online Resource 11).

The two samples of the recently described *H. bolinii* are resolved as sister in both trees (Fig. 2; Online Resource 7) with strong support (ASTRAL = 90 PP; ML = 99 BS). *Hydnora bolinii* is placed sister to *H. solmsiana* in all topologies with strong support (100 PP and BS). The monophyly of Clade I in the ASTRAL tree (Fig. 2) is strongly supported

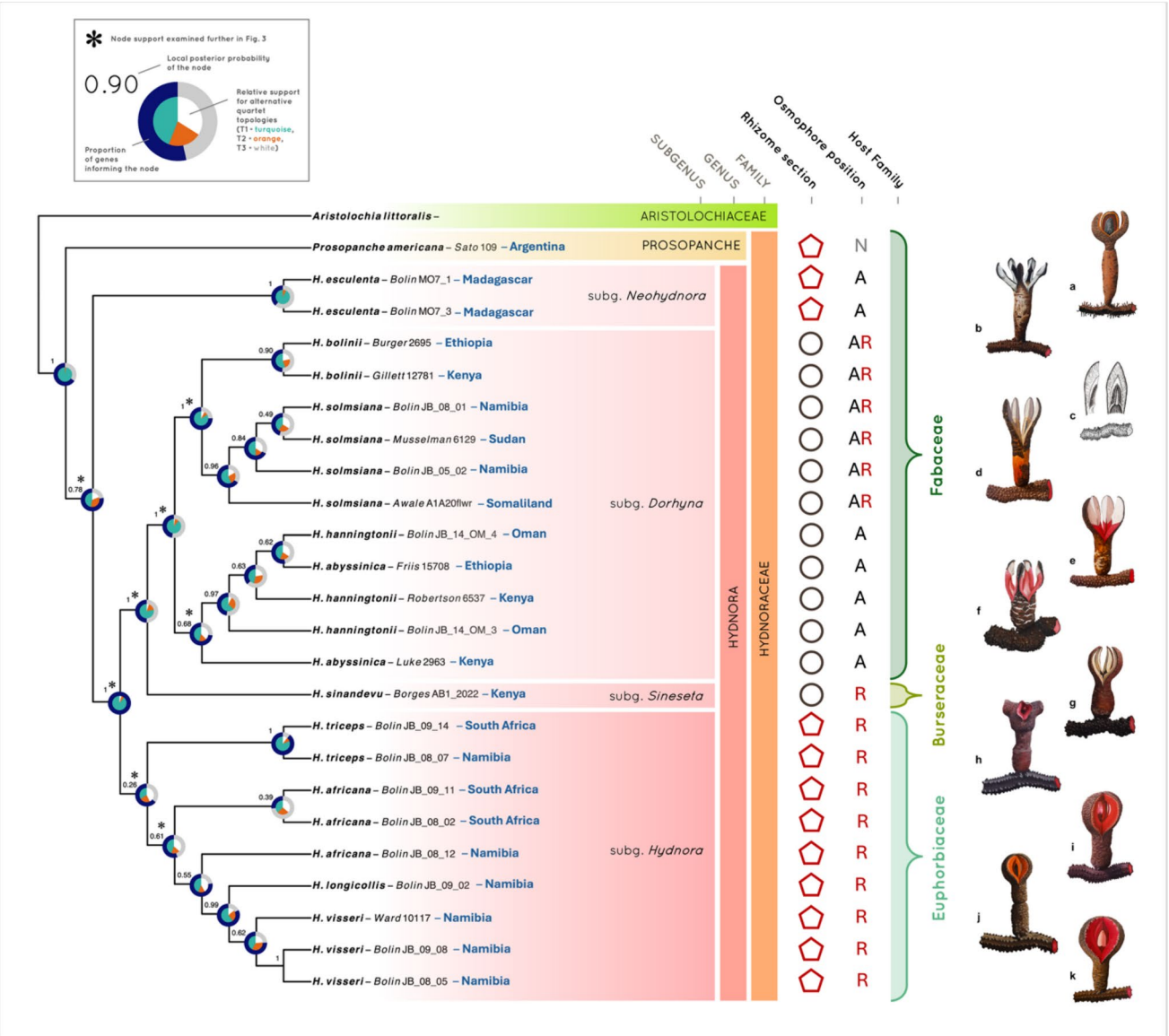


Fig. 2 ASTRAL-III tree based on 34 gene trees, showing relationships within *Hydnora*. For node annotations, the top left number refers to the local posterior probability of the node, the outer pie ring refers to the proportion of genes informing the node, and the inner pie ring refers to the relative support for alternative quartet topologies. **a** *Prosopanche americana*; **b** *Hydnora esculenta*; **c** *H. bolinii*; **d** *H. solmsiana*; **e** *H. hanningtonii*; **f** *H. abyssinica*; **g** *H. sinandevu*; **h** *H. triceps*; **i** *H. visseri*; **j** *H. longicollis*; **k** *H. africana*. **a**–**k** illustrated by Sebastian A. Hatt

by 12 genes and only strongly rejected by one (Fig. 3; Online Resource 11).

Ancestral character state reconstruction

The ancestor of Hydnoraceae is estimated to have angular rhizomes, an apical osmophore but no recessed osmophore, and parasitised Fabaceae (Fig. 4). A single transition to *Euphorbia*-parasitism occurred in the ancestor of subg. *Hydnora*, and a single transition to *Commiphora*-parasitism in the ancestor of subg. *Sineseta* (Fig. 4d). A single transition to terete rhizomes occurred at the ancestor of subg. *Dorhyna* and subg. *Sineseta* (Fig. 4a). Recessed osmophores developed independently in subg. *Hydnora*, subg. *Dorhyna*, and at the ancestor of *H. bolinii* and *H. solmsiana* (Fig. 4b). Apical osmophores were likely lost independently in the ancestor of subg. *Hydnora* and the ancestor of subg. *Sineseta* (Fig. 4c).

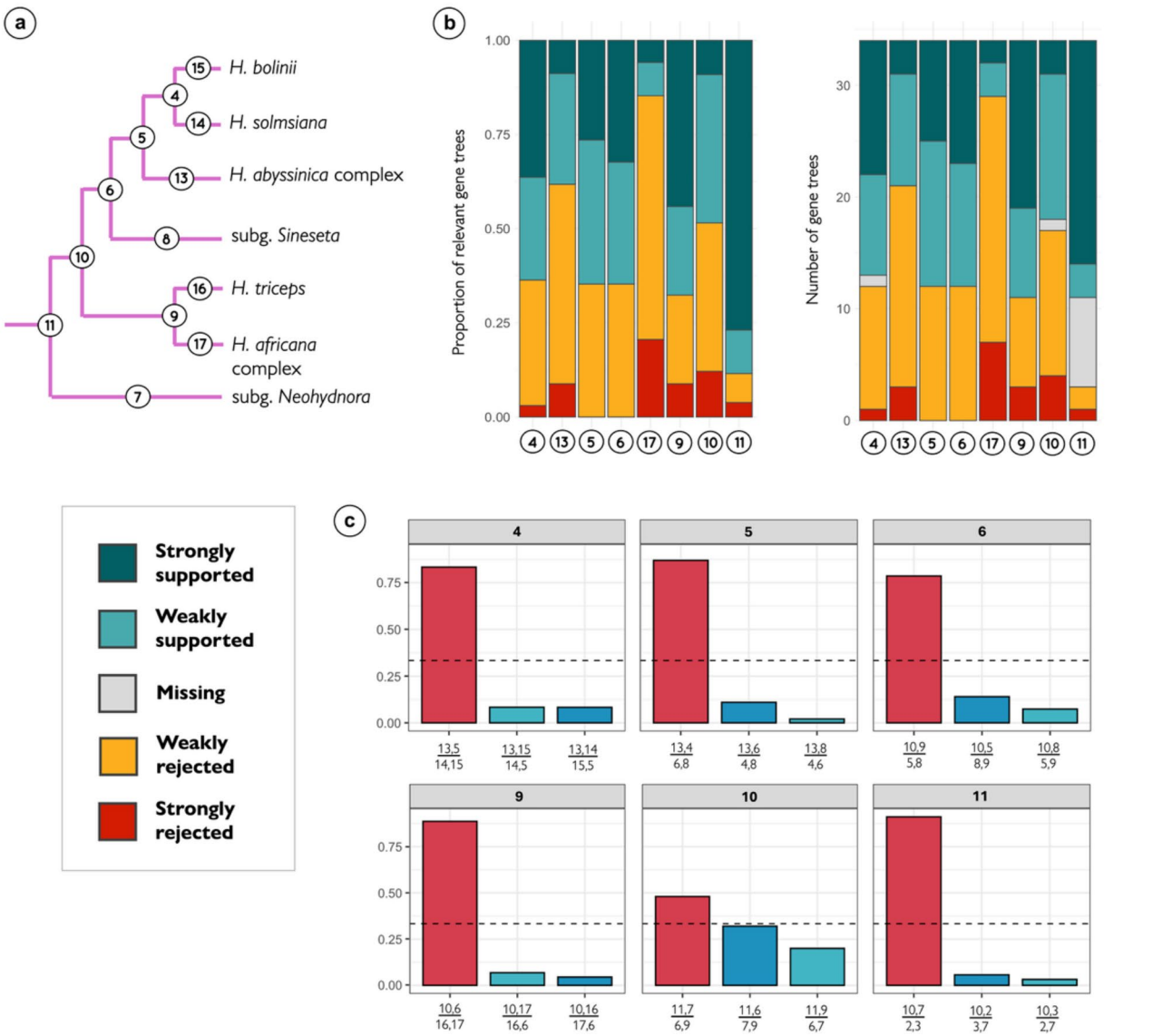


Fig. 3 Distribution of support for gene trees supporting the monophyly of important clades in Fig. 2, generated using DiscoVista. **a** Nodes and branches of interest are numbered and illustrated on a summarised phylogenetic tree; **b** plots summarising the proportion and number of relevant gene trees contributing each level of support for the monophyly of important numbered clades from **a**; **c** plots summarising the relative frequency of alternative topologies at key numbered nodes. The structure of each alternative topology is summarised below the corresponding bar

Discussion

Performance of Angiosperms353 in Hydnoraceae

The derived life history and divergent genomes of parasitic plants have often presented challenges to phylogenetic reconstruction (Nickrent et al. 1998; McNeal et al. 2013; Cai et al. 2026), particularly due to widespread loss of genes associated with photosynthesis (Sun et al. 2018; Gruzdev et al. 2019). Hydnoraceae, a lineage that has likely been parasitic for tens of millions of years (Naumann et al. 2013; Zuntini et al. 2024), has had ample time for such gene losses or for divergences to accumulate (Kim et al. 2025). This may contribute to the low recovery of loci among the 353 targeted by the Angiosperms353 probe set, many of which are nuclear genes associated with photosynthetic pathways (Johnson et al. 2019). Beyond reduced or divergent genomes, low gene recovery may also reflect poor DNA quality. Silica-dried *Hydnora* samples typically showed higher recovery than those from herbarium specimens, suggesting that future efforts to sequence from fresh or silica-dried material of *Prosopanche* may yield higher gene recovery (Online

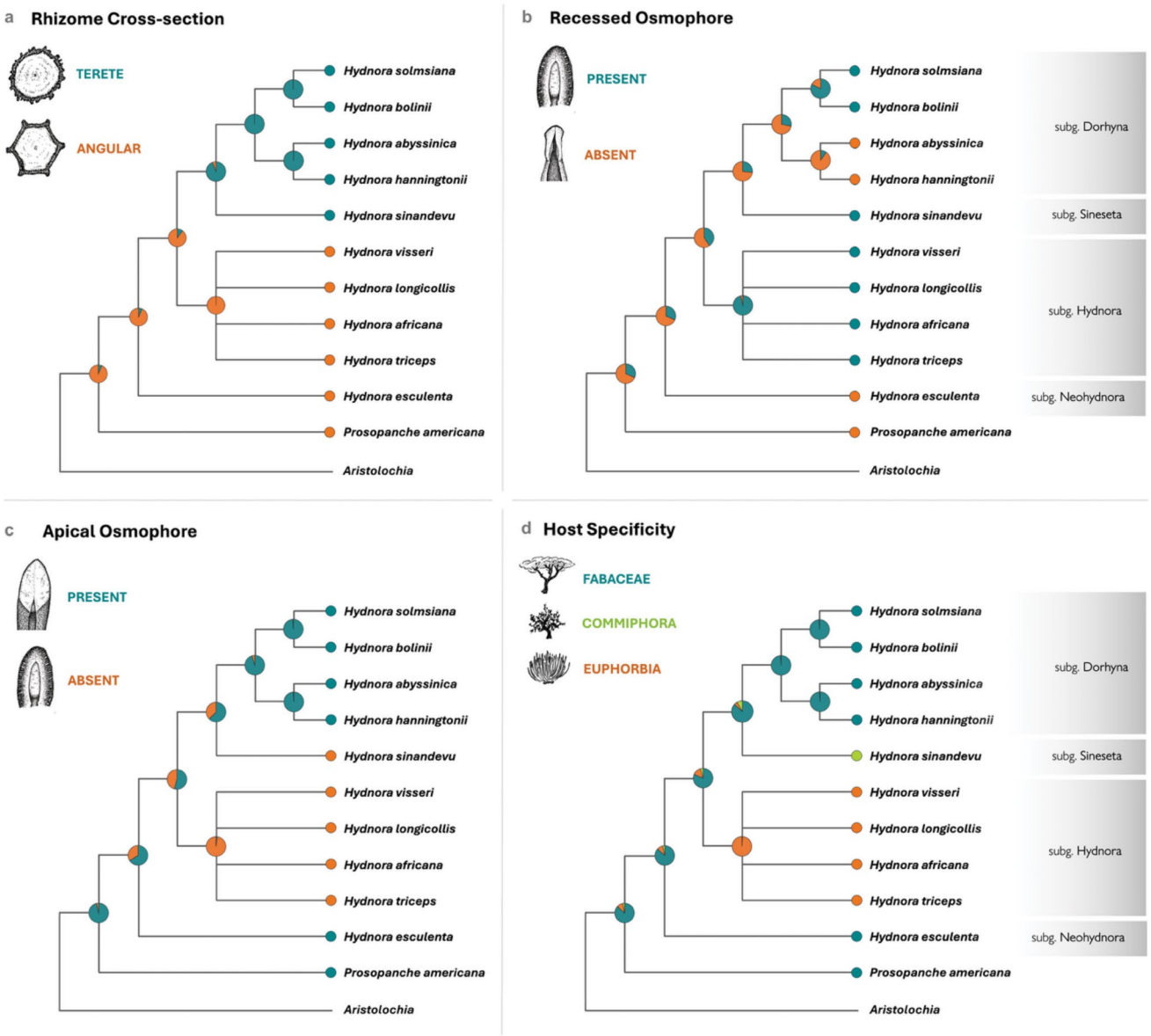


Fig. 4 Ancestral state reconstruction of key taxonomically informative traits in Hydnoraceae, summarised on a simplified maximum likelihood species-level trees: **a** rhizome cross section; **b** recessed osmophore; **c** apical osmophore; **d** host preference. Tips are illustrated with the character state, and nodes are illustrated with the predicted probability of each character state. All drawings by Sebastian Hatt

Resource 10). Despite these limitations, we selected the Angiosperms353 loci because most samples were obtainable only from herbarium specimens, and this kit has been demonstrated to perform well with degraded DNA (Baker et al. 2021). Additionally, it has shown success in recovering genes from a range of holoparasitic taxa in previous studies (Zuntini et al. 2024). Although gene recovery of Hydnoraceae was low compared to the photosynthetic outgroup species, all 34 loci used in the final species-level analysis had were sufficiently polymorphic (Online Resource 4), and the recovered data provided sufficient phylogenetic signal to resolve several major relationships within Hydnoraceae (Fig. 2).

Compared to our study, Zuntini et al. (2024) reported higher gene recovery in Hydnoraceae using the same probe set (239 genes and 63,663 bp for *Hydnora visseri*, versus our recovery of 130 genes/79,616 bp on the same samples). However, direct comparison between these two studies is complicated by methodological differences. In particular, their bioinformatic pipeline (paftools) did not implement a minimum coverage threshold for locus recovery, inflating recovery metrics by including low-confidence or fragmented

sequences. In contrast, our more conservative approach using CAPTUS, which required a minimum depth coverage to retain a locus, prioritises data quality over raw locus count in order to provide a more robust phylogenetic signal.

Strong phylogenomic support for subgeneric classification of Hydnoraceae

Decaisne (1873) proposed floral merosity as a basis for subgeneric classification in *Hydnora*. However, Musselman and Visser (1989) and Hatt et al. (2024b) demonstrated that floral merosity is a highly variable trait and holds little taxonomic utility for infrageneric delimitation. Here, we present a well-sampled nuclear phylogenomic reconstruction that supports the recognition of all four subgenera of Hydnoraceae (Figs. 2, 3; Online Resource 7). Although relationships within some subgenera, particularly subg. *Hydnora*, remain incompletely resolved, the major subgeneric lineages are consistently recovered across analyses. We demonstrate that they are reliably defined by the characters outlined in Hatt et al. (2024b) host preference, the presence/absence and positioning of the apical and recessed osmophore and the cross section of the rhizome (Fig. 4; Table 1). Despite the reduction in morphological characters available for species delimitation in *Hydnora* compared to most angiosperms, this molecular evidence supports the validity of these particular morphological characters for defining infrageneric divisions and species concepts within the genus. The finding is particularly significant for the taxonomy and conservation of parasitic plants more broadly, in which morphological reduction often complicates species identification. By validating these morphological traits with molecular evidence, this study enhances confidence that these traits delimit species, which is essential for accurate biodiversity assessments, prioritising conservation actions, and informing legal protection measures.

Unresolved phylogenetic relationships suggest recent radiation in *Hydnora* subg. *Hydnora*

The exclusively southern African *Hydnora* subgenus *Hydnora* is sister to *H.* subg. *Dorhyna* and *H.* subg. *Sineseta* (Fig. 2, 3) as previously reported (Jost et al. 2022; Mkala et al. 2023), but see Bolin (2009). All members of *Hydnora* subg. *Hydnora* parasitise succulent *Euphorbia* (Euphorbiaceae), each infecting a mutually exclusive set of host species (Hatt et al. 2024b). In agreement with Bolin (2009), our results support a single transition from Fabaceae to *Euphorbia* hosts by an ancestral taxon.

The *Hydnora africana* complex (*H. africana, H. longicollis* and *H. visseri*) comprises an unresolved polytomy of species with flowers that are very similar, differing only in host preference and the relative proportions of the tepals and size of the overall flower (Bolin et al. 2011; Hatt et al. 2024b). Poor gene recovery may have contributed to the lack of phylogenetic resolution in this clade, since two samples of *H. africana* yielded only 29 and 42 genes, respectively, compared to the average recovery of 65 genes across all Hydnoraceae samples. One of these (*Bolin* JB_08_02) was a source of considerable discordance in the SplitsTree analysis (Online Resource 9). Another contributing factor may be the insufficient time for accumulation of mutations in the genes analysed, leading to a lack of informative loci for phylogenetic reconstruction. Indeed, when alignments were filtered to include only sequences from *Hydnora* subg. *Hydnora*, the mean proportion of variable sites was only 0.0565. The hypotheses of recent diversification and flow of genetic material are supported by their mutually exclusive host-parasite relationships with species of *Euphorbia* (Hatt et al. 2024b) and a simultaneous and sympatric radiation that may mirror that of succulent *Euphorbia* across southern Africa in the last 11–2 mya (Horn et al. 2014). A more detailed investigation of this hypothesis would require high-resolution genomic data and robust, time-calibrated phylogenetic trees.

Long-standing taxonomic ambiguity in *Hydnora* subg. *Dorhyna*

Members of *Hydnora* subg. *Dorhyna* have terete rhizomes, parasitise hosts in the Fabaceae, and are distributed from Oman to Nigeria and southern Africa, with the centre of diversity around East Africa. Until the recent taxonomic monograph (Hatt et al. 2024b) divided *H. abyssinica* into three species based on morphological characters of the tepals, any *Hydnora* collected from East Africa was generally attributed to *H. abyssinica* (or the synonymised *H. johannis*). Here, *Hydnora solmsiana* is recovered as a monophyletic species sister to *H. abyssinica* and *H. hanningtonii*; consistent with the results of Mkala et al. (2023). However, the analysis of Mkala et al. (2023) only included a single sample of *H. solmsiana*, limiting its informativeness for species delimitation. Our broader sampling more confidently supports the distinctness of *H. solmsiana* and provides a clearer picture of species boundaries within this clade. *Hydnora abyssinica* and *H. hanningtonii* are morphologically similar, separated only by the colour, proportions and indumentum of the tepals. As with the *H. africana* complex, the poor resolution observed between these two species (Fig. 2) may be explained by recent and ongoing diversification and hybridisation. Given that morphological evidence reliably delimits the two species, the poor molecular resolution should be interpreted cautiously rather than taken necessarily as evidence conspecificity. Future work should combine additional genomic and population-level sampling with morphological and geographic data to better evaluate species boundaries in this clade.

Hydnora bolinii is a recently described species (Hatt et al. 2024b) known only from three specimens, each collected in a different country. Substantial DNA concentrations were successfully recovered from two of these specimens (Online Resource 1), despite both being over 60 years old. Both topologies (Fig. 2; Online Resource 7) resolve the two specimens of *H. bolinii* as a monophyletic species, supporting the species concept of Hatt et al. (2024b). In both topologies, *H. bolinii* resolves as sister to *H. solmsiana*: the species with which *H. bolinii* shares the most morphological characters, namely a terete rhizome and a poorly defined barrier between the tepal margin and the apical osmophore.

Hydnora subgenus *Sineseta*, consisting solely of *H. sinandevu*, is sister to *H.* subg. *Dorhyna* and distinguished by having completely glabrous tepals and parasitising *Commiphora* (Burseraceae). Both the loss of tepal indumentum and transition to *Commiphora*-parasitism are autapomorphies of the subg. *Sineseta* (Fig. 4d).

Ancestral traits of Hydnoraceae

Hydnora esculenta, the sole species of *H.* subgenus *Neohydnora*, is sister to the rest of *Hydnora*, a relationship that has been identified in previous studies using other markers (Bolin 2009; Jost et al. 2021; Mkala et al. 2023). Based on shared character states between *Hydnora* and *Prosopanche*, we postulate that ancestral Hydnoraceae possessed angular rhizomes, apical osmophores and was likely a parasite of Fabaceae (Fig. 4). Rhizomes therefore became terete in the lineage leading to *H.* subg. *Dorhyna* and *H.* subg. *Sineseta*.

The apical osmophore was likely lost independently in *Hydnora* subg. *Hydnora* and *H. sinandevu*. The presence of osmophore tissue in the petals of many Aristolochiaceae (Trujillo and Sérsic 2006; Aliscioni et al. 2017) suggests a common developmental origin with the osmophore tissue in the perianth of Hydnoraceae, although homology is difficult to trace given the extremely divergent morphology of the perianth in Piperales (Fig. 4c). The early-diverging species *H. esculenta* parasitises Fabaceae, a host family preference shared by *H.* subg. *Dorhyna* and several species of *Prosopanche* (Hatt et al. 2023, 2024b). Ancestral state reconstruction (Fig. 4d) therefore suggests the common ancestor of Hydnoraceae likely parasitised Fabaceae, which is known to be a common host family of parasitic plants more widely (Hatt et al. 2024a).

Conclusion

The phylogenetic analyses of Hydnoraceae presented here represent the most comprehensively sampled for the genus *Hydnora* to date. They provide molecular support for the subgeneric classification by Hatt et al. (2024b), which was based entirely on morphological and ecological characters. This provides a valuable backbone for future ecological and evolutionary research into this enigmatic family of parasitic plants. We postulate that ancestral Hydnoraceae had apical osmophores on the perianth, angular rhizomes and parasitised Fabaceae, with transitions to *Euphorbia*-parasitism in subg. *Hydnora* and *Commiphora*-parasitism in subg. *Sineseta*. While some species concepts are strongly supported by molecular evidence for the first time, such as *H. bolinii*, other relationships between and within species are less well supported, particularly in the *H. africana* complex (subg. *Hydnora*) and between *H. abyssinica* and *H. hanningtonii* (subg. *Dorhyna*), where several key nodes show low to moderate support and should therefore not be over-interpreted without additional evidence. Poor gene recovery, recent diversification and hybridisation may contribute to the lack of support for these nodes, and these hypotheses require further testing with expanded genomic datasets and complementary lines of evidence. Population-level sampling with targeted, recently diversified genes will likely be necessary to untangle relationships between and within such species, and should ideally be combined with morphological and geographic data. Future studies working in Hydnoraceae or other holoparasitic or highly specialised lineages with similar challenges could employ transcriptomic or genome-skimming approaches to gain a deeper insight into evolutionary relationships and phylogenetic placement.

Information on Electronic Supplementary Material

Online Resource 1. Samples obtained and used for the phylogenetic analysis.
Online Resource 2. Summary of gene recovery statistics per sample per locus, generated by the CAPTUS extraction phase.
Online Resource 3. Heatmap of recovery (% recovered) per locus (x-axis) for each sample (y-axis) obtained for the phylogenetic analysis.
Online Resource 4. Proportion of variable sites in the species-level gene alignments.
Online Resource 5. Proportion of variable sites and missing data in the species-level gene alignments.
Online Resource 6. The 34 gene trees that contributed to the species tree.
Online Resource 7. Concatenated Maximum Likelihood (ML) species-level tree.
Online Resource 8. Percentage of Hydnoraceae samples containing each gene.
Online Resource 9. Super Network analysis, generated by SplitsTree App, for gene trees contributing to the species tree.
Online Resource 10. Gene recovery per sample type (herbarium specimen vs silica-dried): **a** number of recovered genes; **b** total coding sequence length of recovered genes.
Online Resource 11. Summary of gene support for key clades, generated by DiscoVista gene discordance analysis.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00606-026-02027-7.

Acknowledgements We thank the Royal Botanic Gardens, Kew for funding for the sequencing via the Creative Challenge initiative. Curators of herbaria ALTA, K, L, MO, OBG, RB, TAN and WIND, specimen collectors, and those who provided photographs for Figure 1, are gratefully acknowledged. We thank the HPC Crop Diversity cluster for hosting the servers on which most of the bioinformatic analyses were performed. The silica gel-dried samples were collected by Jay Bolin in 1) Namibia under the Namibian Ministry of Environment and Tourism Research Permit 1013/2006, 2) Oman with the assistance of the Oman Botanic Garden, and 3) Madagascar under the Parcs Nationaux Madagascar Research Permit 291/07.

Author contribution S.A.H. contributed to conceptualisation, investigation, visualisation, writing—original draft, and writing—review and editing. J.F.B., L.J.M., D.D.C., C.J.T., A.R.Z. and O.M.G contributed to writing—review and editing and supervision. J.F.B. contributed to resources. A.M. contributed to investigation.

Funding Royal Botanical Gardens, Kew.

Data availability The data underlying this article are available in the article and in its online Supplementary information. Raw reads are available in ENA (PRJEB XXX [to be filled in on acceptance]).

Declarations

Conflict of interest The authors have no conflict of interests to disclose.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Aliscioni SS, Achler AP, Torretta JP (2017) Floral anatomy, micromorphology and visitor insects in three species of *Aristolochia* L. (Aristolochiaceae). New Zealand J Bot 55:496–513. https://doi.org/10.1080/0028825X.2017.1380051

Baker WJ, Dodsworth S, Forest F, Graham SW, Johnson MG, McDonnell A, Pokorny L, Tate JA, Wicke S, Wickett NJ (2021) Exploring Angiosperms353: an open, community toolkit for collaborative phylogenomic research on flowering plants. Amer J Bot 108:1059–1065. https://doi.org/10.1002/ajb2.1703

Baker WJ, Bailey P, Barber V, Barker A, Bellot S, Bishop D, Botigué LR, Brewer G, Carruthers T, Clarkson JJ, Cook J, Cowan RS, Dodsworth S, Epitawalage N, Françoso E, Gallego B, Johnson MG, Kim JT, Leempoel K, Maurin O, McGinnie C, Pokorny L, Roy S, Stone M, Toledo E, Wickett NJ, Zuntini AR, Eiserhardt WL, Kersey PJ, Leitch IJ, Forest F (2022) A comprehensive phylogenomic platform for exploring the angiosperm tree of life. Syst Biol 71:301–319. https://doi.org/10.1093/sysbio/syab035

Beaulieu JM, O'Meara BC, Donoghue MJ (2013) Identifying hidden rate changes in the evolution of a binary morphological character: the evolution of plant habit in campanulid angiosperms. Syst Biol 62:725–737. https://doi.org/10.1093/sysbio/syt034

Bolin JF, Maass E, Musselman LJ (2009a) Pollination biology of *Hydnora africana* Thunb. (Hydnoraceae) in Namibia: brood-site mimicry with insect imprisonment. Int J Pl Sci 170:157–163. https://doi.org/10.1086/593047

Bolin JF, Maass E, Tennakoon KU, Musselman LJ (2009b) Host-specific germination of the root holoparasite *Hydnora triceps* (Hydnoraceae). Botany 87:1250–1254. https://doi.org/10.1139/B09-078

Bolin JF, Maass E, Musselman LJ (2011) A new species of *Hydnora* (Hydnoraceae) from Southern Africa. Syst Bot 36:255–260. https://doi.org/10.1600/036364411X569453

Bolin JF, Lupton D, Musselman LJ (2018) *Hydnora arabica* (Aristolochiaceae), a new species from the Arabian Peninsula and a key to *Hydnora*. Phytotaxa 338:99. https://doi.org/10.11646/phytotaxa.338.1.8

Bolin JF (2009) Ecology and molecular Phylogenetics of *Hydnora* (Hydnoraceae) in Southern Africa. PhD Thesis, Old Dominion University, Norfolk

Borowiec ML (2016) AMAS: a fast tool for alignment manipulation and computing of summary statistics. PeerJ 4:e1660. https://doi.org/10.7717/peerj.1660

Cai L, Pabón-Mora N, Alzate JF, Liu L, Davis CC (2026) The danger zone: the joint trap of incomplete lineage sorting and long-branch attraction in resolving the Gondwanan origin of Rafflesiaceae and Apodanthaceae. Amer J Bot 113:e70201. https://doi.org/10.1101/2024.08.07.606681

Cocucci AE (1965) Estudios en el género *Prosopanche* (Hydnoraceae), revision taxonómica. Kurtziana 2:53–74

de Pamphilis CW, Palmer JD (1990) Loss of photosynthetic and chlororespiratory genes from the plastid genome of a parasitic flowering plant. Nature 348:337–339. https://doi.org/10.1038/348337a0

de Carvalho JDT, Markus C, Junior AM, Záchia RA, Schuch M, Sulzbach E, Mariath JEA (2021) *Prosopanche cocuccii* (Hydnoraceae): a new species from Southern Brazil. Phytotaxa 521:177–192. https://doi.org/10.11646/phytotaxa.521.3.3

Doyle JJ, Doyle JL (1987) A rapid DNA isolation procedure for small quantities of fresh leaf tissue. Phytochem Bull 19:11–15

ESRI (2026) ArcGIS Pro Version 3.2. Environmental Systems Resource Institute, Redlands

Funez LA, Ribeiro-Nardes W, Kossmann T, Peroni N, Drechsler-Santos ER (2019) *Prosopanche demogorgoni*: a new species of *Prosopanche* (Aristolochiaceae: Hydnoroideae) from southern Brazil. Phytotaxa 422:93–100. https://doi.org/10.11646/phytotaxa.422.1.6

GBIF.org (2025) GBIF Home Page. Available at: www.gbif.org. Accessed 2 Sep 2025

Gruzdev EV, Kadnikov VV, Beletsky AV, Mardanov AV, Ravin NV (2019) Extensive plastome reduction and loss of photosynthesis genes in *Diphelypaea coccinea*, a holoparasitic plant of the family Orobanchaceae. PeerJ 7:e7830. https://doi.org/10.7717/peerj.7830

Hatt SA, Cameron DD, Grace OM, Rocamundi N, Cocucci AA, Martel C, Thorogood CJ (2023) *Prosopanche*: a remarkable genus of parasitic plants. Pl People Planet 5:163–168. https://doi.org/10.1002/ppp3.10340

Hatt SA, Grace OM, Zuntini AR, Cameron DD, Thorogood CJ (2024a) Parasitic plants show striking convergence in host preference across angiosperm lineages. Ann Bot 135:1135–1146. https://doi.org/10.1093/aob/mcae180

Hatt SA, Thorogood CJ, Bolin JF, Musselman LJ, Cameron DD, Grace OM (2024b) A taxonomic revision of the genus *Hydnora* (Hydnoraceae). Kew Bull 79:459–514. https://doi.org/10.1007/s12225-024-10193-5

Helmstetter AJ, Ezedin Z, de Lírio EJ, de Oliveira SM, Chatrou LW, Erkens RHJ, Larridon I, Leempoel K, Maurin O, Roy S,

Zuntini AR, Baker WJ, Couvreur TLP, Forest F, Sauquet H (2025) Toward a phylogenomic classification of magnoliids. Amer J Bot 112:e16451. https://doi.org/10.1002/ajb2.16451

Horn JW, Xi Z, Riina R, Peirson JA, Yang Y, Dorsey BL, Berry PE, Davis CC, Wurdack KJ (2014) Evolutionary bursts in *Euphorbia* (Euphorbiaceae) are linked with photosynthetic pathway: evolutionary bursts in *Euphorbia*. Evolution 68:3485–3504. https://doi.org/10.1111/evo.12534

Huson DH, Bryant D (2024) The SplitsTree app: interactive analysis and visualization using phylogenetic trees and networks. Nature Meth 21:1773–1774. https://doi.org/10.1038/s41592-024-02406-3

Johnson MG, Pokorny L, Dodsworth S, Botigué LR, Cowan RS, Devault A, Eiserhardt WL, Epitawalage N, Forest F, Kim JT, Leebens-Mack J, Maurin O, Soltis DE, Soltis PS, Wurdack KJ, Baker WJ (2019) A universal probe set for targeted sequencing of 353 nuclear genes from any flowering plant designed using k-medoids clustering. Syst Biol 68:594–606. https://doi.org/10.1093/sysbio/syy086

Jost M, Samain M-S, Marques I, Graham SW, Wanke S (2021) Discordant phylogenomic placement of Hydnoraceae and Lactoridaceae within Piperales using data from all three genomes. Frontiers Pl Sci 12:642598. https://doi.org/10.3389/fpls.2021.642598

Jost M, Naumann J, Bolin JF, Martel C, Rocamundi N, Cocucci AA, Lupton D, Neinhuis C, Wanke S (2022) Structural plastome evolution in holoparasitic Hydnoraceae with special focus on inverted and direct repeats. Genome Biol Evol 14:evac077. https://doi.org/10.1093/gbe/evac077

Junier T, Zdobnov EM (2010) The Newick utilities: high-throughput phylogenetic tree processing in the Unix shell. Bioinformatics 26:1669–1670. https://doi.org/10.1093/bioinformatics/btq243

Katoh K, Standley DM (2013) MAFFT multiple sequence alignment software version 7: improvements in performance and usability. Molec Biol Evol 30:772–780. https://doi.org/10.1093/molbev/mst010

Keller O, Odronitz F, Stanke M, Kollmar M, Waack S (2008) Scipio: using protein sequences to determine the precise exon/intron structures of genes and their orthologs in closely related species. BMC Bioinf 9:278. https://doi.org/10.1186/1471-2105-9-278

Kim W, Schmidt N, Jost M, Mkala EM, Winkler S, Hu G, Heitkam T, Wanke S (2025) Diverging repeatomes in holoparasitic Hydnoraceae uncover a playground of genome evolution. New Phytol 247:1520–1537. https://doi.org/10.1111/nph.70280

Kim W, Jost M, Nickrent D, Zhou R, Acar P, Langschied F, Ebersberger I, Wicke S, Wanke S (2026) Progress and prospects of parasitic plant biodiversity genomics. Pl Cell Physiol 67:438–459. https://doi.org/10.1093/pcp/pcag009

Kozlov AM, Darriba D, Flouri T, Morel B, Stamatakis A (2019) RAxML-NG: a fast, scalable and user-friendly tool for maximum likelihood phylogenetic inference. Bioinformatics 35:4453–4455. https://doi.org/10.1093/bioinformatics/btz305

Li D, Liu C-M, Luo R, Sadakane K, Lam T-W (2015) MEGAHIT: an ultra-fast single-node solution for large and complex metagenomics assembly via succinct de Bruijn graph. Bioinformatics 31:1674–1676. https://doi.org/10.1093/bioinformatics/btv033

Mai U, Mirarab S (2018) TreeShrink: fast and accurate detection of outlier long branches in collections of phylogenetic trees. BMC Genomics 19:272. https://doi.org/10.1186/s12864-018-4620-2

Martel C, Fernandez-Hilario R, Tello JA, Arteaga RG, Gerlach G (2018) *Prosopanche panguanensis* (Aristolochiaceae), a new species from central Peru. Phytotaxa 364:241. https://doi.org/10.11646/phytotaxa.364.3.3

Massoni J, Forest F, Sauquet H (2014) Increased sampling of both genes and taxa improves resolution of phylogenetic relationships within Magnoliidae, a large and early-diverging clade of angiosperms. Molec Phylogen Evol 70:84–93. https://doi.org/10.1016/j.ympev.2013.09.010

McNeal JR, Bennett JR, Wolfe AD, Mathews S (2013) Phylogeny and origins of holoparasitism in Orobanchaceae. Amer J Bot 100:971–983. https://doi.org/10.3732/ajb.1200448

Meyer E (1833) De Hydnora. Nov Acta Phys-Med Acad Caes Leop-Carol Nat Cur 16:784–785

Mkala EM, Mutungi MM, Mutinda ES, Omondi SF, Musili PM, Mwachala G, Hu G-W, Wanke S, Wang Q-F (2021) Understanding the Ethnobotany, Chemistry, Pharmacology, and Distribution of Genus *Hydnora* (Aristolochiaceae). Plants 10:494. https://doi.org/10.3390/plants10030494

Mkala EM, Jost M, Dong X, Mwachala G, Musili PM, Wanke S, Hu G-W, Wang Q-F (2023) Phylogenetic and comparative analyses of *Hydnora abyssinica* plastomes provide evidence for hidden diversity within Hydnoraceae. BMC Ecol Evol 23:34. https://doi.org/10.1186/s12862-023-02142-w

Musselman L, Visser J (1989) Taxonomy and natural history of *Hydnora* (Hydnoraceae). Aliso 12:317–326. https://doi.org/10.5642/aliso.19891202.09

Naumann J, Salomo K, Der JP, Wafula EK, Bolin JF, Maass E, Frenzke L, Samain M-S, Neinhuis C, de Pamphilis CW, Wanke S (2013) Single-copy nuclear genes place haustorial Hydnoraceae within Piperales and reveal a Cretaceous origin of multiple parasitic angiosperm lineages. PLoS ONE 8:e79204. https://doi.org/10.1371/journal.pone.0079204

Nickrent DL, Duff RJ, Colwell AE, Wolfe AD, Young ND, Steiner KE, dePamphilis CW (1998) Molecular phylogenetic and evolutionary studies of parasitic plants. In: Soltis DE, Soltis PS, Doyle JJ (eds) Molecular systematics of plants II. Springer US, Boston, pp 211–241

Nickrent DL, Blarer A, Qiu Y-L, Vidal-Russell R, Anderson FE (2002) Molecular data place Hydnoraceae with Aristolochiaceae. Amer J Bot 89:1809–1817. https://doi.org/10.3732/ajb.89.11.1809

Nickrent DL, Blarer A, Qiu Y-L, Vidal-Russell R, Anderson FE (2004) Phylogenetic inference in Rafflesiales: the influence of rate heterogeneity and horizontal gene transfer. BMC Evol Biol 4:40. https://doi.org/10.1186/1471-2148-4-40

Ortiz EM, Höwener A, Shigita G, Raza M, Maurin O, Zuntini A, Forest F, Baker WJ, Schaefer H (2023) A novel phylogenomics pipeline reveals complex pattern of reticulate evolution in Cucurbitales. Bioinformatics. https://doi.org/10.1101/2023.10.27.564367

Paradis E, Schliep K (2019) Ape 5.0: an environment for modern phylogenetics and evolutionary analyses in R. Bioinformatics 35:526–528. https://doi.org/10.1093/bioinformatics/bty633

Percival-Alwyn L, Barnes I, Clark MD, Cockram J, Coffey MP, Jones S, Kersey PJ, Kidner CA, Kosiol C, Li B, Marsh WA, Zhou J, Caccamo M, Milne I (2024) UKCropDiversity-HPC: a collaborative high-performance computing resource approach for sustainable agriculture and biodiversity conservation. Pl People Planet 7:969–977. https://doi.org/10.1002/ppp3.10607

Rocamundi N, Arce Miller M, Maubecin CC, Martel C, Moré M, Marvaldi A, Cocucci AA (2024) While *Prosopanche* (Hydnoraceae) flowers gently heat: mutualistic pollination relationships among the perianth-bearing Piperales. Bot J Linn Soc 204:199–211. https://doi.org/10.1093/botlinnean/boad050

Sayyari E, Whitfield JB, Mirarab S (2018) DiscoviSta: interpretable visualizations of gene tree discordance. Molec Phylogen Evol 122:110–115. https://doi.org/10.1016/j.ympev.2018.01.019

Sun G, Xu Y, Liu H, Sun T, Zhang J, Hettenhausen C, Shen G, Zhang X, Li J, Guo Y, Liu M, Zhuang H, Wang X, Li Y, Wu J, Song J, Liu X, Ma X, Wu Z, Zhang Y, Qin Y, Long CL, Li J (2018) Large-scale gene losses underlie the genome evolution of parasitic plant *Cuscuta australis*. Nat Commun 9:2683. https://doi.org/10.1038/s41467-018-04721-8

Teixeira-Costa L, Heberling JM, Wilson CA, Davis CC (2023) Parasitic flowering plant collections embody the extended specimen. Meth Ecol Evol 14:319–331. https://doi.org/10.1111/2041-210x.13866

Tennakoon KU, Bolin JF, Musselman LJ, Maass E (2007) Structural attributes of the hypogeous holoparasite *Hydnora triceps* Drège & Meyer (Hydnoraceae). Amer J Bot 94:1439–1449. https://doi.org/10.3732/ajb.94.9.1439

Trujillo CG, Sérsic AN (2006) Floral biology of *Aristolochia argentina* (Aristolochiaceae). Flora 201:374–382. https://doi.org/10.1016/j.flora.2005.07.013

Williams VL, Falcão MP, Wojtasik EM (2011) *Hydnora abyssinica*: Ethnobotanical evidence for its occurrence in southern Mozambique. S African J Bot 77:474–478. https://doi.org/10.1016/j.sajb.2010.09.010

Yu G, Smith DK, Zhu H, Guan Y, Lam TT-Y (2017) Ggtree: an R package for visualization and annotation of phylogenetic trees with their covariates and other associated data. Meth Ecol Evol 8:28–36. https://doi.org/10.1111/2041-210X.12628

Zhang C, Rabiee M, Sayyari E, Mirarab S (2018) ASTRAL-III: polynomial time species tree reconstruction from partially resolved gene trees. BMC Bioinf 19:153. https://doi.org/10.1186/s12859-018-2129-y

Zuntini AR, Carruthers T, Maurin O, Bailey PC, Leempoel K, Brewer GE, Epitawalage N, Françoso E, Gallego-Paramo B, McGinnie C, Negrão R, Roy SR, Simpson L, Toledo Romero E, Barber VMA, Botigué L, Clarkson JJ, Cowan RS, Dodsworth S, Johnson MG, Kim JT, Pokorny L, Wickett NJ, Antar GM, DeBolt L, Gutierrez K, Hendriks KP, Hoewener A, Hu A-Q, Joyce EM, Kikuchi IABS, Larridon I, Larson DA, de Lírio EJ, Liu J-X, Malakasi P, Przelomska NAS, Shah T, Viruel J, Allnutt TR, Ameka GK, Andrew RL, Appelhans MS, Arista M, Ariza MJ, Arroyo J, Arthan W (2024) Phylogenomics and the rise of the angiosperms. Nature 629:843–850. https://doi.org/10.1038/s41586-024-07324-0

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.